



ASIAN INSTITUTE OF MANAGEMENT
Executive Education Program

PREDICTING STUDENT DROPOUT AND ACADEMIC SUCCESS
USING MACHINE LEARNING TECHNIQUES

A Final Narrative Report

Submitted in Partial Fulfillment
of the Requirements for the Completion
of the course Post Graduate Diploma in Artificial
Intelligence and Machine Learning

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INTRODUCTION

Student retention and academic success remain among the most significant challenges faced by higher education institutions worldwide. Every year, a considerable number of students discontinue their studies due to various academic, financial, personal, and socioeconomic factors. Student dropout not only affects the individuals involved by limiting their educational and career opportunities but also impacts institutional performance, resource allocation, and national educational development. As educational institutions increasingly adopt data-driven decision-making, machine learning has emerged as a valuable tool for identifying students who may be at risk of dropping out, enabling timely interventions that can improve student outcomes.

This final narrative report presents the complete implementation of a machine learning project entitled "Predicting Student Dropout and Academic Success Using Machine Learning Techniques." The project utilizes the publicly available Students Dropout and Academic Success dataset to develop predictive models capable of classifying students into three academic outcomes: Dropout, Enrolled, and Graduate. By analyzing demographic, socioeconomic, financial, and academic variables, the study aims to uncover the key factors influencing student success and build an accurate predictive framework that can support institutional decision-making.

The project follows the standard data science lifecycle, beginning with problem understanding and framing, followed by data collection and understanding, data preprocessing, exploratory data analysis (EDA), and feature engineering. Multiple supervised machine learning algorithms—including Logistic Regression, Decision Tree, Random Forest, and XGBoost—were implemented and evaluated to determine the most effective predictive model. The study also incorporates feature importance analysis, model explainability, and ethical AI practices, including fairness and bias auditing, to ensure that the resulting models are not only accurate but also transparent, interpretable, and responsible.

Throughout the project, appropriate evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix were used to compare model performance. In addition, explainable AI techniques, including SHAP and LIME, were considered to provide meaningful interpretations of model predictions and to support informed educational decision-making. Ethical considerations, fairness assessments, and bias mitigation strategies were also explored to ensure that the predictive models promote equitable treatment of students from diverse backgrounds.

This report demonstrates how machine learning can be effectively applied to educational data to identify students who may require additional academic or financial support before they disengage from their studies. The findings provide valuable insights for educators, administrators, and policymakers seeking to improve student retention, enhance graduation rates, and develop evidence-based intervention programs. By combining robust data analytics with responsible AI principles, this

project contributes to the growing role of predictive analytics in creating more inclusive, proactive, and student-centered educational environments.

Chapter 1

Understanding and Framing

The objective of this project is to analyze the Students Dropout and Academic Success dataset to identify the factors that influence whether a student successfully graduates, remains enrolled, or drops out. Student attrition is a significant challenge for educational institutions because it affects academic performance indicators, institutional planning, and the efficient allocation of educational resources. By understanding the characteristics associated with student outcomes, schools can implement timely interventions that improve student retention and academic success.

1.1 Business Problem

Student retention and academic success are important concerns for educational institutions because student dropout can affect academic performance indicators, institutional planning, resource allocation, and overall educational effectiveness. Identifying students who are likely to graduate, remain enrolled, or discontinue their studies can provide institutions with information that may support timely and targeted interventions.

The Students Dropout and Academic Success dataset provides demographic, socioeconomic, admission, financial, and academic information that can be used to examine factors associated with different student outcomes. The dataset contains 4,424 student records and 37 variables, including information on previous qualifications, admission performance, family background, scholarship status, tuition payment status, and first- and second-semester academic performance.

The exploratory analysis indicates that academic performance is particularly relevant to student outcomes. Variables related to the number of curricular units approved and semester grades demonstrated strong relationships with whether students graduated or dropped out. Financial indicators, including tuition payment status and scholarship status, also showed meaningful associations with student outcomes. These findings suggest that educational institutions may benefit from a predictive approach that uses available student information to identify patterns associated with academic success and potential dropout.

The central business problem addressed in this study is therefore the need to improve the identification of students who may be at risk of dropping out and to support timely, evidence-based interventions that can improve student retention and academic success. A predictive decision-support approach may assist educators and administrators in prioritizing appropriate support such as academic advising, mentoring, counseling, tutoring, and financial assistance.

1.2 Data Science Problem

From a data science perspective, the problem involves developing a predictive machine learning model capable of classifying students according to their academic outcome using historical student data. The dataset contains demographic, socioeconomic, financial, admission, and academic-performance variables that collectively provide information about the student's academic trajectory.

The target variable, Target, contains three possible outcomes: Dropout, Enrolled, and Graduate. Dropout represents students who discontinued their studies before completion, Enrolled represents students who remained actively enrolled, and Graduate represents students who successfully completed their academic program.

The exploratory analysis also identified a moderate difference in the distribution of the three target classes, with Graduate students representing the largest group and Enrolled students the smallest group. This characteristic is important because a model could achieve relatively high overall accuracy while providing weaker predictions for a smaller class. Consequently, the data science problem requires an evaluation approach that considers both overall and class-specific predictive performance.

The predictive model is intended not only to classify student outcomes but also to provide insights into the factors associated with these predictions. The subsequent feature-importance analysis identified academic performance variables, particularly first- and second-semester curricular units and grades, together with tuition status, scholarship status, admission grade, and age at enrollment, among the important predictors of student outcomes.

1.3 Machine Learning Task

This study is formulated as a supervised multiclass classification problem.

The target variable is Target, representing the student's academic outcome. The target consists of three mutually exclusive classes:

Dropout – Students who discontinued their studies before completing their academic program.

Enrolled – Students who remain actively enrolled in their academic program.

Graduate – Students who successfully completed their academic program.

Multiclass classification is appropriate because the objective is to predict one of three predefined categorical outcomes using historical student data with known outcomes. Regression techniques are unsuitable because the target variable is categorical rather than continuous, while unsupervised clustering would not directly address the objective of predicting known academic outcomes.

The project evaluates several supervised classification algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The comparison of these models is intended to determine which approach provides the most useful

predictive performance for distinguishing among Dropout, Enrolled, and Graduate students.

1.4 Success Metrics

The predictive performance of the machine learning models will be evaluated using several complementary classification metrics to provide a comprehensive assessment of model effectiveness.

Accuracy will measure the overall proportion of students correctly classified into their respective academic outcome categories. Precision will assess the proportion of students predicted to belong to a particular class who actually belong to that class, while Recall will measure the model's ability to correctly identify students who actually belong to each outcome category.

F1-Score, which represents the harmonic mean of Precision and Recall, will provide a balanced measure of classification performance. Particular attention will be given to the macro-average F1-score because the target classes are not equally represented in the dataset. This metric gives equal consideration to Dropout, Enrolled, and Graduate rather than allowing the largest class to dominate the overall assessment.

The Confusion Matrix will also be analyzed to identify the specific types of classification errors produced by each model. This is particularly important for determining whether students who actually drop out are being incorrectly classified as Enrolled or Graduate, as such errors could represent missed opportunities for early intervention.

1.5 Business Key Performance Indicators

The practical value of the predictive model will be assessed through business-oriented and institutional Key Performance Indicators that reflect potential improvements in student outcomes and educational effectiveness.

The primary institutional objective is to improve student retention by enabling educational institutions to identify students who may be at greater risk of dropping out. Earlier identification can provide educators and administrators with an opportunity to implement appropriate interventions before academic difficulties result in student withdrawal.

Successful application of the predictive model may also contribute to improved early intervention effectiveness by helping institutions prioritize students who may require academic advising, tutoring, mentoring, counseling, or financial support. The model may further contribute to more efficient allocation of student-support resources, allowing limited institutional resources to be directed toward students with greater identified needs.

Another important KPI is the graduation or academic success rate. If students identified as potentially at risk receive effective and timely support, the institution may ultimately achieve improved graduation outcomes. These institutional outcomes should be regarded as downstream measures of the potential value of the predictive model and would require actual implementation and longitudinal evaluation to establish measurable impact.

1.6 Capstone Deliverable

The primary deliverable for this phase is a clearly defined machine learning problem that connects the institutional challenge of student retention and academic success with predictive analytics. Specifically, this phase establishes the business problem, formulates the corresponding data science problem, identifies the machine learning task as supervised multiclass classification, defines the target variable as Target, and establishes the technical success metrics of Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis.

The problem definition is also grounded in the actual characteristics of the dataset and the results of the exploratory and predictive analyses. The analysis identified academic performance, particularly curricular-unit completion and semester grades, as important factors associated with student outcomes, together with financial and admission-related characteristics.

This problem framing provides the foundation for the subsequent stages of data understanding, data quality assessment, exploratory data analysis, preprocessing, categorical encoding, feature engineering, model development, model evaluation, and explainable AI analysis.

Chapter 2

Data Collection and Understanding

2.1 Data Collection

The dataset used in this study is the Students Dropout and Academic Success dataset (<https://www.kaggle.com/datasets/adilshamim8/predict-students-dropout-and-academic-success>), which is publicly available through the UCI Machine Learning Repository and distributed through Kaggle. It was selected because it contains comprehensive demographic, socioeconomic, financial, admission, and academic information that is relevant to understanding and predicting student academic outcomes. The dataset provides information on students' previous qualifications, admission performance, family background, financial circumstances, and academic achievements during the first and second semesters. These characteristics provide a broad representation of the factors that may be associated with a student's educational trajectory.

The dataset consists of 4,424 student records and 37 variables, comprising 36 predictor variables and one target variable. The predictor variables capture different aspects of a student's profile, including admission information, family background, financial status, demographic characteristics, and academic performance. The target variable, Target, classifies each student into one of three academic outcome categories: Dropout, Enrolled, and Graduate.

Since each student record has a known academic outcome, the dataset is suitable for a supervised multiclass classification problem. The available predictor variables provide the information required to train machine learning models to distinguish among the three target categories.

Dataset Overview

Attribute	Description
Dataset Name	Students Dropout and Academic Success
Source	UCI Machine Learning Repository / Kaggle
Number of Records	4,424
Number of Variables	37 (36 predictors and 1 target)
Target Variable	Target
Target Classes	Dropout, Enrolled, Graduate
Problem Type	Supervised Multiclass Classification

2.2 Data Understanding

Initial exploration of the dataset was conducted to examine its structure, data quality, and variable characteristics before proceeding to data preprocessing and model development. The dataset contains a combination of categorical, binary, ordinal, and numerical variables, allowing different dimensions of student characteristics and academic performance to be represented. Categorical and binary variables describe demographic, institutional, and financial characteristics, while ordinal variables represent characteristics with an inherent ordering. Numerical variables provide quantitative measures of academic performance, admission characteristics, age, and broader economic conditions.

The categorical variables include characteristics such as gender, marital status, course, scholarship status, debtor status, tuition payment status, and international student status. The ordinal variables include Mother's Qualification, Father's Qualification, Mother's Occupation, Father's Occupation, and Application Order. Numerical variables include Admission Grade, Previous Qualification Grade, Age at Enrollment, Curricular Units Enrolled, Curricular Units Approved, semester grades, Unemployment Rate, Inflation Rate, and GDP.

The combination of these variables provides a multidimensional representation of the student population. In particular, the inclusion of both academic and financial characteristics allows the study to examine whether student outcomes are associated

not only with academic performance but also with financial and socioeconomic circumstances.

2.3 Data Quality Assessment

A data quality assessment was conducted to determine whether the dataset contained issues that could affect subsequent analysis and model development. The assessment considered missing values, duplicate records, data types, and potential outliers.

The initial inspection confirmed that the dataset contains no missing values across the 4,424 student records. As a result, no data imputation techniques were required. This reduced the preprocessing requirements and allowed the analysis to focus on the appropriate representation of categorical variables, numerical feature scaling, and preparation of the dataset for machine learning.

Duplicate-record analysis also showed that there were no duplicate records in the dataset. Each observation was therefore treated as a unique student record, eliminating the need for duplicate removal before further analysis.

Outlier detection was performed on the numerical variables using descriptive statistics and visualization techniques, including boxplots. Several variables, including Age at Enrollment, Admission Grade, Previous Qualification Grade, and Curricular Units, exhibited extreme observations. However, these observations were assessed as legitimate student records rather than obvious data-entry errors. Consequently, the observations were retained because removing them could potentially eliminate meaningful information about student characteristics and academic performance.

The overall data quality assessment indicates that the dataset is structurally complete and requires relatively limited cleaning before modeling. The primary preprocessing requirements therefore involve appropriate treatment of the different feature types, categorical encoding, and scaling or normalization of numerical variables where required by the selected machine learning algorithms.

2.4 Feature Types

The dataset contains several types of predictor variables, each representing a different aspect of student characteristics and academic performance.

Information about the DataFrame (data types, non-null values, memory usage):

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4424 entries, 0 to 4423
```

```
Data columns (total 37 columns):
```

#	Column	Non-Null Count	Dtype
0	Marital Status	4424 non-null	int64
1	Application mode	4424 non-null	int64
2	Application order	4424 non-null	int64
3	Course	4424 non-null	int64
4	Daytime/evening attendance	4424 non-null	int64
5	Previous qualification	4424 non-null	int64
6	Previous qualification (grade)	4424 non-null	float64
7	Nacionality	4424 non-null	int64
8	Mother's qualification	4424 non-null	int64
9	Father's qualification	4424 non-null	int64
10	Mother's occupation	4424 non-null	int64
11	Father's occupation	4424 non-null	int64
12	Admission grade	4424 non-null	float64
13	Displaced	4424 non-null	int64
14	Educational special needs	4424 non-null	int64
15	Debtor	4424 non-null	int64
16	Tuition fees up to date	4424 non-null	int64
17	Gender	4424 non-null	int64
18	Scholarship holder	4424 non-null	int64
19	Age at enrollment	4424 non-null	int64
20	International	4424 non-null	int64
21	Curricular units 1st sem (credited)	4424 non-null	int64
22	Curricular units 1st sem (enrolled)	4424 non-null	int64
23	Curricular units 1st sem (evaluations)	4424 non-null	int64
24	Curricular units 1st sem (approved)	4424 non-null	int64
25	Curricular units 1st sem (grade)	4424 non-null	float64
26	Curricular units 1st sem (without evaluations)	4424 non-null	int64
27	Curricular units 2nd sem (credited)	4424 non-null	int64
28	Curricular units 2nd sem (enrolled)	4424 non-null	int64
29	Curricular units 2nd sem (evaluations)	4424 non-null	int64
30	Curricular units 2nd sem (approved)	4424 non-null	int64
31	Curricular units 2nd sem (grade)	4424 non-null	float64
32	Curricular units 2nd sem (without evaluations)	4424 non-null	int64
33	Unemployment rate	4424 non-null	float64
34	Inflation rate	4424 non-null	float64
35	GDP	4424 non-null	float64
36	target	4424 non-null	object

```
dtypes: float64(7), int64(29), object(1)
```

The categorical features include Gender, Marital Status, Course, Scholarship Holder, Debtor, Tuition Fees Up to Date, International Student, and Target. These variables represent demographic, institutional, financial, and outcome-related characteristics.

The ordinal features include Mother's Qualification, Father's Qualification, Mother's Occupation, Father's Occupation, and Application Order. These variables

contain categories that represent an order or ranking and therefore require appropriate treatment during the preprocessing stage.

The numerical features include Admission Grade, Previous Qualification Grade, Age at Enrollment, Curricular Units Enrolled, Curricular Units Approved, semester grades, Unemployment Rate, Inflation Rate, and GDP. These variables provide quantitative information about students' academic performance, admission characteristics, age, and the broader economic environment.

The combination of demographic, academic, financial, and economic variables provides a broad feature space for modeling student academic outcomes. It also allows the subsequent analysis to examine which characteristics have the strongest relationship with whether students Dropout, remain Enrolled, or Graduate.

2.5 Target Variable

The target variable, Target, represents the academic outcome of each student and contains three categories: Dropout, Enrolled, and Graduate. These categories form the response variable for the supervised machine learning models developed in this study.

Initial exploratory analysis showed that the three target categories were not equally represented.

```
Unique values and their counts for the 'target' column:
target
Graduate      2209
Dropout       1421
Enrolled       794
Name: count, dtype: int64
```

Graduate students constituted the largest proportion of the dataset, while Enrolled students represented the smallest group. This indicates a moderate degree of class imbalance that must be considered during model development and evaluation.

This characteristic is important because a model could achieve relatively high overall accuracy by performing well on the largest class while providing weaker predictions for the smaller Enrolled class. Consequently, model evaluation will

consider not only overall Accuracy but also class-specific Precision, Recall, F1-Score, and Confusion Matrix results.

2.6 Initial Data Patterns

The initial exploration of the dataset revealed several meaningful relationships between predictor variables and student outcomes. Academic performance variables demonstrated particularly strong associations with the target variable. In particular, the number of curricular units approved during the first and second semesters showed strong relationships with semester grades and student outcomes.

Students who successfully completed a greater number of curricular units generally demonstrated a higher probability of graduating, while students with fewer approved units were more likely to drop out. This finding provides an important basis for the subsequent predictive modeling because it suggests that early academic performance contains substantial information about a student's eventual academic outcome.

Financial characteristics also demonstrated meaningful relationships with student outcomes. Scholarship recipients showed higher graduation rates than non-scholarship students. Similarly, students whose tuition fees were up to date exhibited considerably lower dropout rates than students with outstanding financial obligations. Debtor status also showed an association with academic success.

These findings suggest that student academic outcomes are influenced by a combination of academic performance and financial circumstances. They also provide an empirical basis for including these variables in the subsequent machine learning analysis rather than treating all predictors as equally informative.

2.7 Data Dictionary

Feature	Data Type	Description
Marital Status	int64	Categorical: Marital status of the student (e....
Application mode	int64	Categorical: Mode of application (e.g., 1st ph...

Application order	int64	Ordinal: Order of application (e.g., 1st, 2nd,...
Course	int64	Categorical: Course code. Each number correspo...
Daytime/evening attendance	int64	Binary: 1 for daytime attendance, 0 for evenin...
Previous qualification	int64	Categorical: Previous academic qualification (...)
Previous qualification (grade)	float64	Numerical: Grade obtained in previous academic...
Nacionality	int64	Categorical: Nationality of the student. Speci...
Mother's qualification	int64	Categorical: Qualification of the student's mo...
Father's qualification	int64	Categorical: Qualification of the student's fa...
Mother's occupation	int64	Categorical: Occupation of the student's mothe...
Father's occupation	int64	Categorical: Occupation of the student's fathe...
Admission grade	float64	Numerical: Grade obtained in the admission pro...
Displaced	int64	Binary: Whether the student is displaced (1 if...
Educational special needs	int64	Binary: Whether the student has special educat...
Debtor	int64	Binary: Whether the student is a debtor (1 if ...)
Tuition fees up to date	int64	Binary: Whether the student has paid tuition f...
Gender	int64	Binary: Gender of the student (e.g., 1 for mal...
Scholarship holder	int64	Binary: Whether the student is a scholarship h...
Age at enrollment	int64	Numerical: Age of the student at the time of e...
International	int64	Binary: Whether the student is an internationa...
Curricular units 1st sem (credited)	int64	Numerical: Number of curricular units credited...
Curricular units 1st sem (enrolled)	int64	Numerical: Number of curricular units enrolled...
Curricular units 1st sem (evaluations)	int64	Numerical: Number of evaluations performed in ...
Curricular units 1st sem (approved)	int64	Numerical: Number of curricular units approved...
Curricular units 1st sem (grade)	float64	Numerical: Average grade obtained in the 1st s...
Curricular units 1st sem (without evaluations)	int64	Numerical: Number of curricular units in the 1...
Curricular units 2nd sem (credited)	int64	Numerical: Number of curricular units credited...
Curricular units 2nd sem (enrolled)	int64	Numerical: Number of curricular units enrolled...
Curricular units 2nd sem (evaluations)	int64	Numerical: Number of evaluations performed in ...
Curricular units 2nd sem (approved)	int64	Numerical: Number of curricular units approved...

Curricular units 2nd sem (grade)	float64	Numerical: Average grade obtained in the 2nd s...
Curricular units 2nd sem (without evaluations)	int64	Numerical: Number of curricular units in the 2...
Unemployment rate	float64	Numerical: Unemployment rate in the period of ...
Inflation rate	float64	Numerical: Inflation rate in the period of enr...
GDP	float64	Numerical: Gross Domestic Product in the perio...
target	object	Categorical: The academic outcome of the stude...

2.8 Chapter Summary

Overall, the Students Dropout and Academic Success dataset is a complete and structurally suitable dataset for the multiclass classification problem. It contains 4,424 student records and 37 variables, with no missing values and no duplicate records. Although several numerical variables contained outliers, these observations were considered valid student records and were retained for subsequent analysis.

The initial data understanding also revealed important characteristics that inform the succeeding stages of the project. The target variable contains three outcome categories with a moderate imbalance, while exploratory analysis identified meaningful relationships between academic performance, financial circumstances, and student outcomes. In particular, curricular-unit completion, semester grades, tuition status, and scholarship status demonstrated notable associations with academic outcomes.

Chapter 3

Data Preprocessing, Applied Exploratory Data Analysis (EDA), and Feature Engineering

This chapter presents the data preparation, exploratory analysis, and feature engineering procedures applied to the Students Dropout and Academic Success dataset. The objective of these procedures was to assess the quality and structure of the data, address potential issues that could affect model performance, transform the variables into a suitable machine-learning representation, and identify the features most relevant to predicting student academic outcomes.

The dataset contains 4,424 observations and 37 variables, with the target variable representing three student outcomes: Dropout, Enrolled, and Graduate. The preprocessing and exploratory procedures were conducted prior to model development to establish a consistent and suitable dataset for the subsequent classification models.

3.1 Data Preprocessing

Data preprocessing began with an assessment of the dataset's structure and quality. The variables were examined according to their data types, completeness, and suitability for subsequent statistical and machine-learning analysis. The initial inspection showed that the dataset contained 4,424 observations and 37 variables and that no missing values were present. Consequently, no missing-value imputation was necessary.

3.1.1 Missing-Value Assessment

A missing-value assessment was performed across the dataset. The analysis confirmed that the observations contained complete values for the variables used in the analysis. Since no missing observations were identified, procedures such as mean, median, or mode imputation were not required. The absence of missing values simplified the preprocessing stage and allowed the analysis to proceed without introducing imputation-related assumptions or additional sources of variability.

3.1.2 Outlier Detection and Treatment

Outlier analysis was conducted on the numerical variables using descriptive statistics and boxplots. The numerical variables examined for potential extreme observations included Previous Qualification (grade), Admission Grade, Curricular Units 1st Semester (grade), Curricular Units 2nd Semester (grade), Unemployment Rate, Inflation Rate, and GDP.

Rather than deleting observations identified as extreme, a percentile-based capping procedure was applied. For each identified numerical variable, values below the 1st percentile were replaced by the 1st-percentile value, while values above the 99th percentile were replaced by the 99th-percentile value.

This approach preserved all student observations while limiting the potential influence of extreme numerical values on subsequent analyses. It was particularly appropriate because the extreme observations were part of the observed student data rather than being automatically treated as erroneous records. Thus, the preprocessing did not remove potentially meaningful student observations. Instead, it reduced the influence of extreme numerical values while maintaining the original sample size of 4,424 observations.

3.1.3 Categorical Variable Encoding

Several predictor variables were categorical or represented categorical information using numerical codes. To convert these variables into a machine-readable representation, one-hot encoding was applied using the `pandas.get_dummies()` function with `drop_first=True`.

One-hot encoding transformed categorical variables into binary indicator variables. The `drop_first=True` option removed one reference category from each encoded categorical variable, thereby reducing redundant information and avoiding unnecessary perfect multicollinearity among dummy variables.

This transformation allowed categorical information, such as student demographic, academic, and financial characteristics, to be incorporated into the machine-learning models without treating category labels as continuous numerical measurements.

3.1.4 Feature Scaling

Numerical features were standardized using StandardScaler as part of the preprocessing procedure. Standardization transformed numerical variables to a common scale based on their mean and standard deviation.

This step was important because the dataset contains variables measured using different numerical scales, including grades, age, curricular-unit counts, and macroeconomic indicators. Standardization provided a consistent numerical representation for subsequent statistical analysis and modeling.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine the statistical characteristics of the dataset and identify patterns and relationships between predictor variables and student academic outcomes.

The EDA employed both numerical summaries and graphical techniques. These included descriptive statistics, histograms, boxplots, categorical frequency visualizations, and correlation analysis. The results provided an initial understanding of the distribution of the variables and the characteristics associated with the three target classes.

3.2.1 Target Variable Distribution

The target variable represents the student's academic outcome and consists of three categories: Dropout, Enrolled, and Graduate.

The distribution showed that Graduate was the largest class, followed by Dropout, while Enrolled represented the smallest class. The resulting class distribution is therefore not perfectly balanced, making class-specific evaluation important during model assessment.

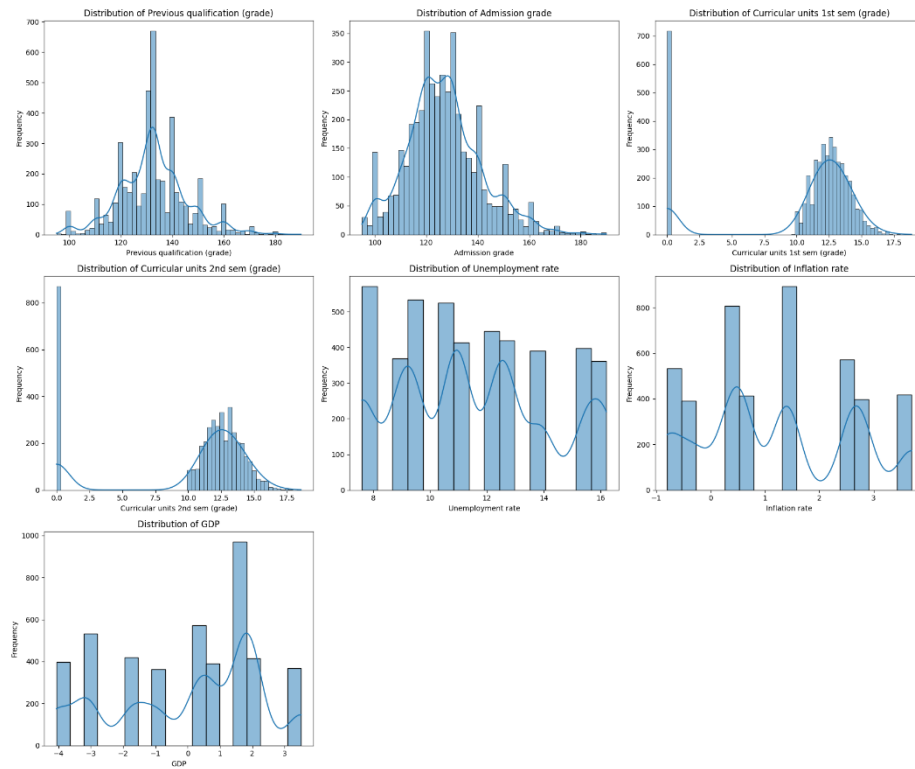
This characteristic is particularly relevant because a model can achieve relatively high overall accuracy while performing poorly on a smaller class. Consequently, the project evaluated the models using not only accuracy but also precision, recall, and F1-score for the individual classes.

3.2.2 Descriptive Statistics and Numerical Distributions

Descriptive statistics were generated for the numerical variables to examine their central tendency, dispersion, and observed ranges. Histograms and density plots were also used to visualize the distributions of numerical variables.

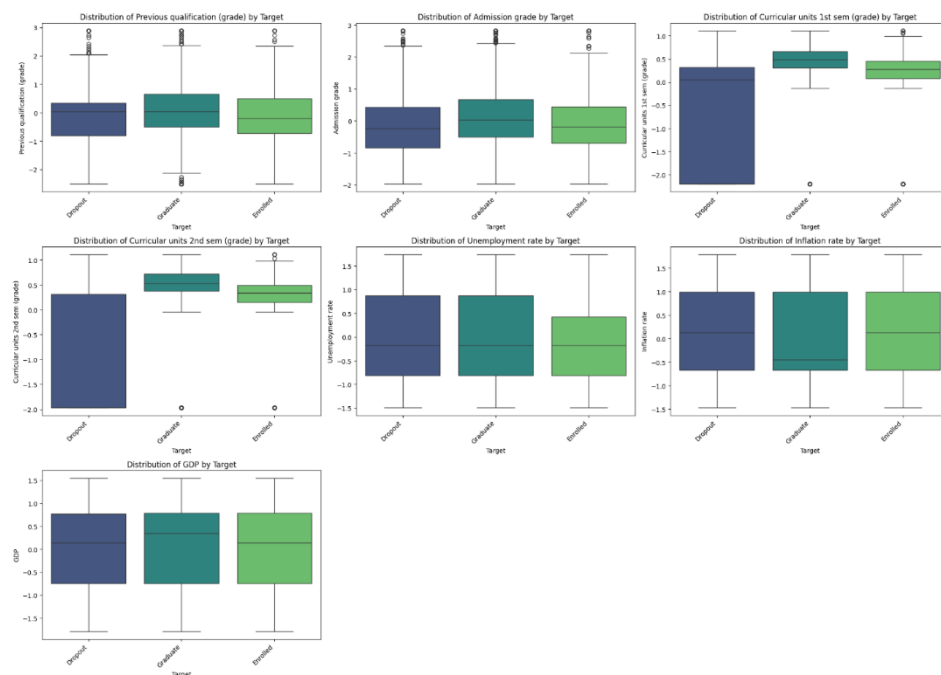
The analysis provided a basis for understanding the characteristics of variables such as admission grade, previous qualification grade, semester grades, and macroeconomic indicators. These visualizations were also useful in identifying variables with potentially skewed distributions and extreme observations requiring further examination.

The descriptive statistics following percentile-based capping showed, for example, that Previous Qualification (grade) had a mean of approximately 132.57, while Admission Grade had a mean of approximately 126.93 after the treatment.



3.2.3 Outlier Visualization

Boxplots were used to visually examine the numerical variables for extreme observations. The boxplots provided an initial basis for identifying variables requiring outlier treatment.

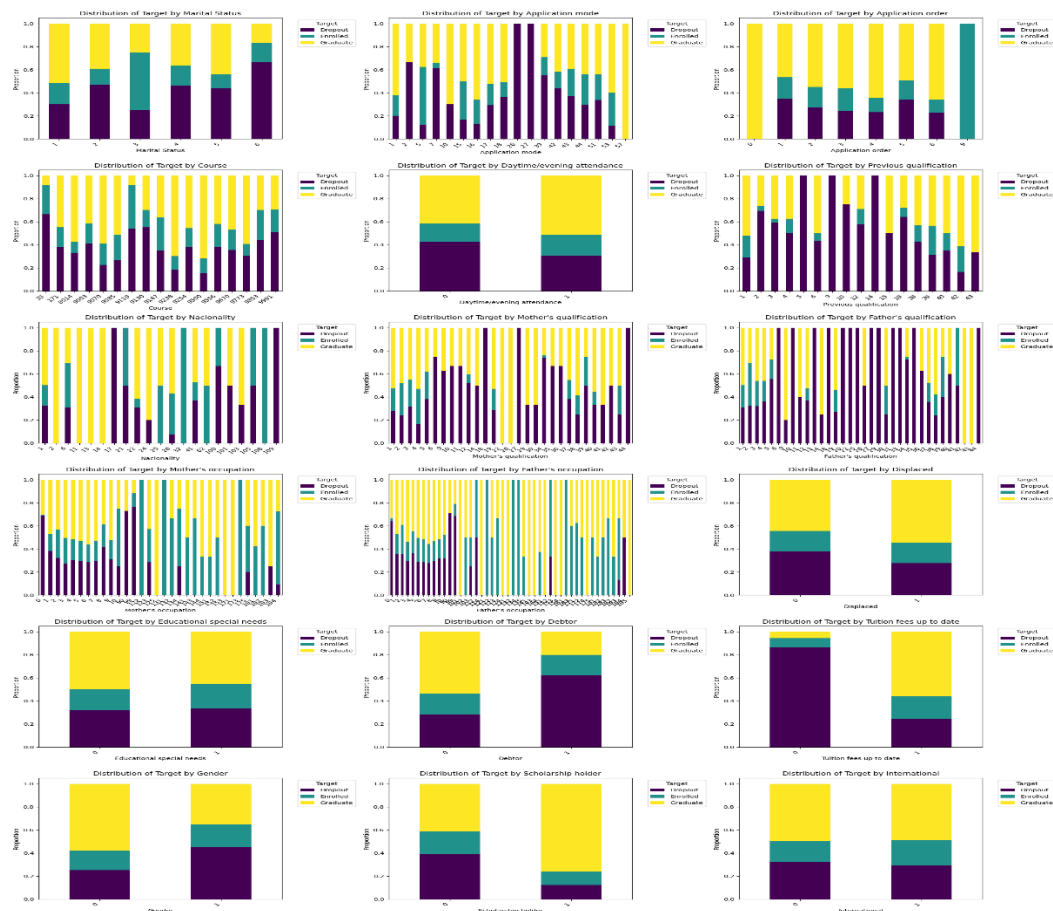


Following the identification of extreme values, the selected numerical variables were subjected to 1st–99th percentile capping rather than observation removal. This maintained the complete dataset while reducing the influence of extreme values.

3.2.4 Analysis of Categorical Variables

Categorical variables were examined using frequency distributions and graphical comparisons with the target variable. These analyses provided an initial understanding of how demographic, academic, and financial characteristics varied across the three student outcomes.

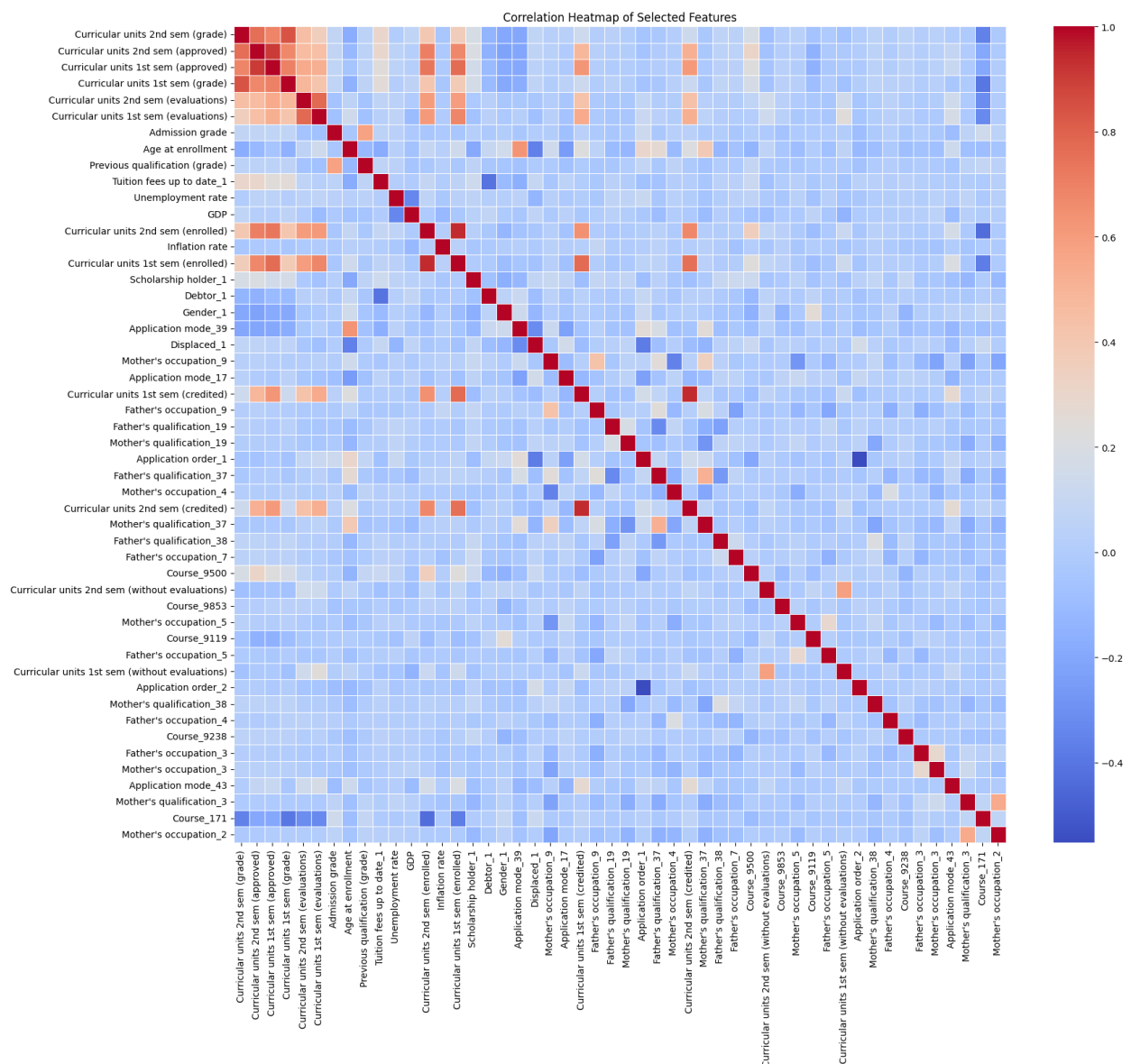
The EDA particularly highlighted the importance of variables related to student academic performance and financial status. The subsequent machine-learning analysis provided a more systematic assessment of these relationships through model-based feature importance and explainability techniques.



3.2.5 Correlation Analysis

Correlation analysis was performed to examine relationships among numerical variables. The resulting correlation matrix provided insight into the degree of linear association among academic and other numerical characteristics.

Strong relationships were observed among several academic-performance variables. In particular, variables associated with curricular-unit performance and semester grades demonstrated substantial relationships with one another. These relationships are substantively meaningful because successful completion and performance in curricular units are directly related to a student's academic progression.



The correlation analysis also informed the subsequent feature-selection process by providing an understanding of potential redundancy among predictor variables.

3.3 Feature Engineering and Transformation

Feature engineering in this study focused primarily on transforming the existing variables into representations appropriate for machine-learning algorithms rather than creating numerous new domain-specific variables.

The principal transformations were outlier treatment, categorical encoding, numerical scaling, and feature selection. These procedures preserved the information contained in the original dataset while producing a machine-readable feature matrix.

3.3.1 Numerical Feature Standardization

Numerical variables were standardized using `StandardScaler`. This transformation placed numerical variables on a comparable scale and prevented differences in measurement units from unnecessarily affecting algorithms that are sensitive to feature magnitude.

The standardized numerical variables were subsequently incorporated with the encoded categorical variables to form the processed predictor matrix.

3.3.2 One-Hot Encoding of Categorical Variables

Categorical predictors were transformed through one-hot encoding with the first category dropped as the reference category. This procedure converted categorical information into binary indicator variables while avoiding redundant dummy variables.

The result was a higher-dimensional numerical feature matrix that could be directly used by the classification algorithms.

3.3.3 Construction of the Modeling Feature Matrix

After preprocessing and encoding, the transformed variables were combined into a unified feature matrix. Feature importance analysis was subsequently used to identify the most informative predictors.

The notebook selected the top 50 features based on model-derived feature importance before proceeding to the final modeling stage. This produced `X_selected`, which served as the principal feature matrix used for model development. The subsequent train-test split consisted of 3,539 training observations and 885 testing observations across the 50 selected features.

3.4 Applied Exploratory Data Analysis

To provide a comprehensive understanding of the dataset, several visualization techniques were applied during the EDA phase.

These included histograms and density plots for examining numerical-variable distributions; boxplots for identifying potential extreme observations; bar charts for examining categorical variables across target classes; correlation heatmaps for assessing relationships among numerical variables; and feature-importance visualizations for identifying the variables contributing most strongly to the classification task.

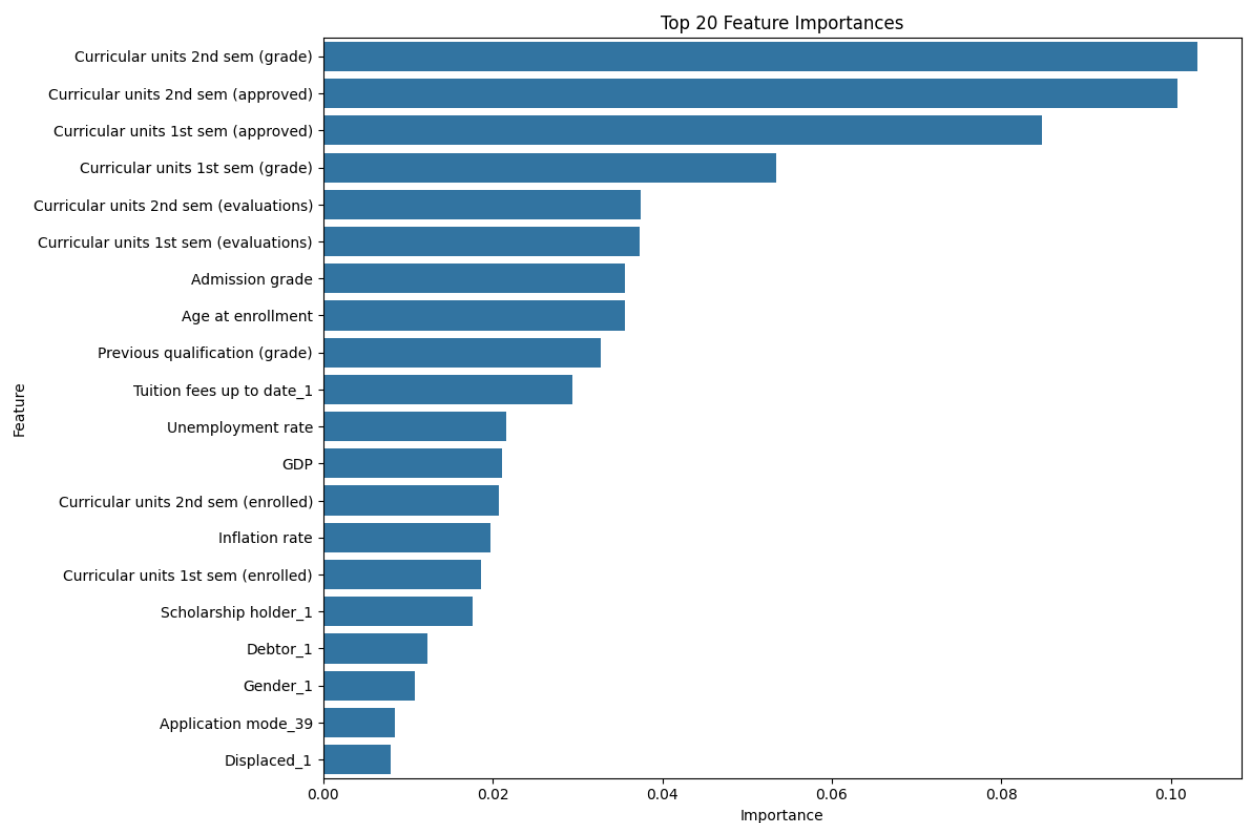
These visualizations complemented the numerical summaries and provided a more comprehensive understanding of the dataset.

In particular, the EDA indicated that academic-performance variables were strongly associated with the student outcomes. This observation was subsequently supported by the Random Forest feature-importance analysis and the explainability analyses conducted later in the project.

3.5 Feature Importance

Feature importance analysis was conducted to identify the predictors that contributed most strongly to the classification of students into Dropout, Enrolled, and Graduate categories.

A Random Forest Classifier was used to obtain model-based feature importance scores. These scores provided a quantitative basis for ranking the predictor variables according to their contribution to the classification model.



The analysis identified academic-performance variables among the most influential predictors. In particular, variables associated with curricular-unit grades and performance during the first and second semesters, together with admission grade, previous qualification grade, tuition-fee status, and age at enrollment, emerged as important predictors.

These findings indicate that the student's academic performance, particularly during the early stages of tertiary education, provides substantial information for distinguishing among the three academic outcomes. This finding is consistent with the subsequent SHAP and partial-dependence analyses, which provided more detailed explanations of how the important variables influenced model predictions.

3.6 Feature Selection

Feature selection was performed to reduce the dimensionality of the predictor space and retain the variables that provided the greatest predictive information.

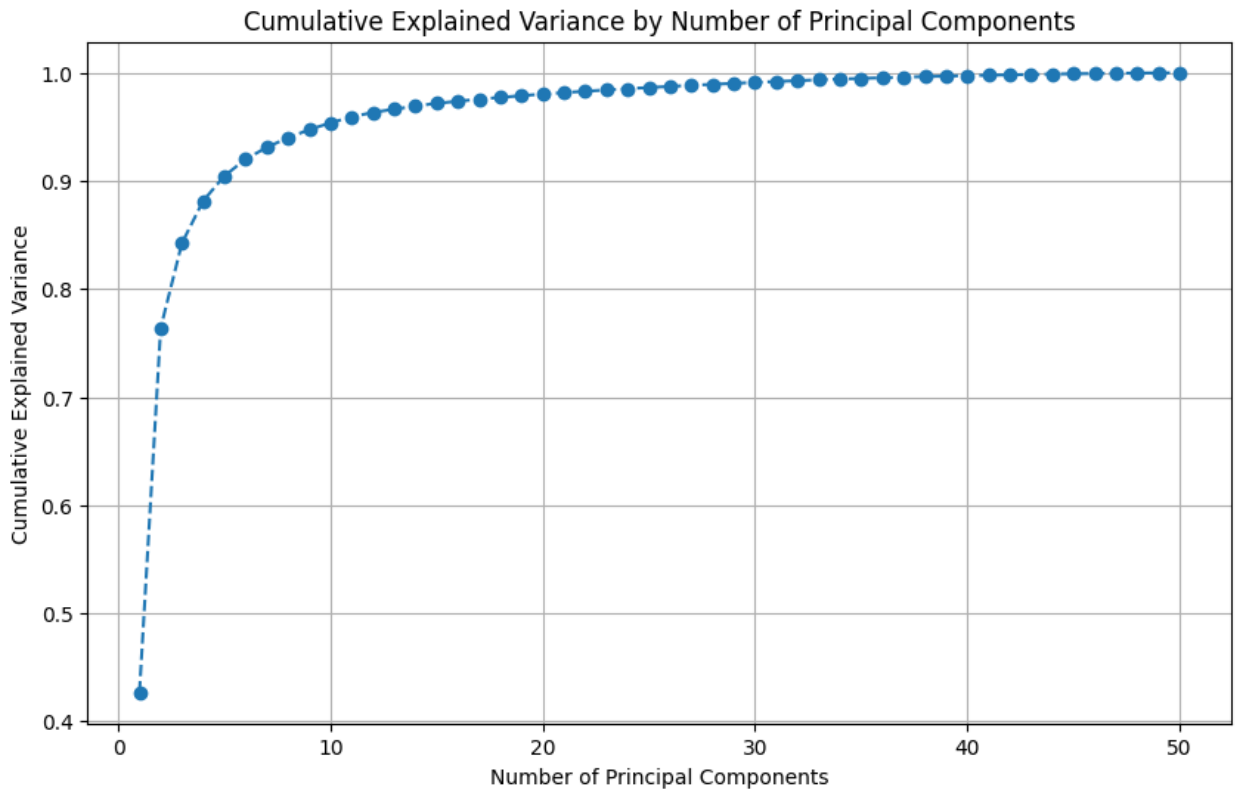
Following preprocessing and encoding, Random Forest feature importance was used to rank the transformed predictors. The 50 highest-ranked features were selected for subsequent modeling.

This approach allowed the analysis to retain the most informative predictors while reducing the number of variables supplied to the classification algorithms. The resulting feature matrix contained 50 predictors and was used for the subsequent train-test split and model development.

Feature selection also provided a basis for subsequent model interpretation because the explainability analyses were conducted using the selected feature representation.

3.7 Principal Component Analysis

Principal Component Analysis (PCA) was conducted as an additional dimensionality-reduction analysis. PCA was fitted to the selected feature matrix to examine the underlying structure of the transformed data and determine how the variance was distributed across principal components.



The PCA analysis provided an alternative representation of the selected features and allowed the project to examine whether a smaller number of orthogonal components could capture substantial information contained in the original feature space.

Importantly, PCA was used as an analytical dimensionality-reduction procedure and was not used as the final feature representation for the classification models. The final models were trained using the selected feature matrix (X_{selected}), which contained the 50 features identified through feature-importance-based selection. This distinction is important in accurately describing the modeling pipeline.

3.8 Data Preparation for Model Development

Following preprocessing, feature transformation, feature selection, and PCA analysis, the selected feature matrix was divided into training and testing datasets.

An 80:20 stratified train-test split was applied using `random_state=42`. Stratification ensured that the distribution of the three target classes was maintained across the training and testing datasets. The resulting datasets consisted of a training

set with 3,539 observations and 50 features and a testing set with 885 observations and 50 features.

This resulted in an independent test set that could be used to evaluate the predictive performance of the classification models.

3.9 Summary of Data Preprocessing, EDA, and Feature Engineering

The data preparation phase established the foundation for the subsequent machine-learning analysis. The dataset contained 4,424 observations and 37 variables and did not require missing-value imputation. Numerical variables were examined for extreme observations, and selected numerical features were treated using 1st–99th percentile capping rather than removing observations.

Categorical predictors were transformed through one-hot encoding, while numerical variables were standardized. Exploratory analysis using descriptive statistics and graphical techniques provided insight into variable distributions and relationships among student characteristics and academic outcomes.

Random Forest feature importance was then used to identify the most informative predictors, resulting in the selection of the top 50 features for model development. PCA was additionally performed to examine the dimensional structure of the selected feature space, but the PCA representation was not used as the final input to the classification models.

Finally, the selected features were divided using an 80:20 stratified train-test split, producing 3,539 training observations and 885 testing observations.

Chapter 4

Model Implementation

This chapter presents the development, training, and evaluation of machine learning models for predicting student academic outcomes using the Students Dropout and Academic Success dataset. The objective was to determine whether the available demographic, financial, and academic characteristics could be used to classify students into one of three outcomes: Dropout, Enrolled, or Graduate.

Following the data preprocessing, exploratory analysis, and feature-selection procedures described in Chapter 3, the resulting dataset was prepared for supervised multiclass classification. Four machine learning algorithms were implemented and compared: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The models were evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis.

The purpose of the model comparison was not simply to identify the model with the highest overall accuracy, but to determine which model provided the most balanced predictive performance across the three student outcome categories.

4.1 Model Development and Implementation

The modeling dataset consisted of the 50 features selected through the Random Forest feature-importance procedure described in Chapter 3. The selected feature matrix was divided into training and testing subsets using an 80:20 stratified split. The resulting training dataset contained 3,539 observations, while the testing dataset contained 885 observations. Stratification was used to maintain the representation of the three target classes across the training and testing subsets.

The four classification algorithms implemented in the notebook represent different approaches to supervised multiclass classification. Logistic Regression provided a relatively simple linear baseline, Decision Tree provided an interpretable rule-based approach, while Random Forest and XGBoost represented ensemble-learning approaches capable of modeling more complex relationships.

4.1.1 Logistic Regression

Logistic Regression was implemented as a baseline classification model. The model provides a relatively simple approach to multiclass prediction and establishes a reference point against which the more complex algorithms can be evaluated.

The Logistic Regression model achieved an overall test accuracy of approximately 0.77 and a macro F1-score of approximately 0.69. Its class-specific performance was strongest for the Graduate and Dropout categories, while prediction of the Enrolled category remained more challenging.

4.1.2 Decision Tree

A Decision Tree classifier was implemented to model relationships between predictors and student outcomes through a series of hierarchical decision rules.

Among the four models evaluated, Decision Tree produced the lowest overall predictive performance. It achieved an accuracy of approximately 0.66 and a macro F1-score of approximately 0.60. This result indicates that a single-tree approach was less effective than the ensemble models for capturing the complexity of the student outcome classification problem.

4.1.3 Random Forest

Random Forest was implemented as an ensemble classifier consisting of multiple decision trees. By aggregating the predictions of multiple trees, Random Forest can reduce the variance associated with individual decision trees and provide more robust classification performance.

The Random Forest model achieved an overall test accuracy of approximately 0.77 and the highest macro F1-score among the evaluated models at approximately 0.70. Its class-specific F1-scores were approximately 0.78 for Dropout, 0.47 for Enrolled, and 0.85 for Graduate.

These results indicate that Random Forest provided the strongest balance of predictive performance across the three target categories.

4.1.4 XGBoost

XGBoost was implemented as a gradient-boosting ensemble model. The algorithm builds a sequence of decision trees in which subsequent trees are designed to improve upon errors made by previous trees.

XGBoost achieved an overall accuracy of approximately 0.76 and a macro F1-score of approximately 0.69. Its performance was therefore competitive with Logistic Regression and Random Forest, particularly for the Graduate and Dropout classes.

However, based on the reported evaluation metrics, XGBoost did not achieve the highest overall performance in this project. Random Forest achieved the highest macro F1-score and slightly higher overall accuracy.

4.2 Model Evaluation

The models were evaluated using the independent testing dataset. Since the objective was to classify students into three distinct academic outcomes, evaluation considered both overall and class-specific performance.

Accuracy measures the proportion of all test observations that were correctly classified. Precision measures the proportion of observations predicted as belonging to a particular class that actually belong to that class. Recall measures the proportion of actual observations belonging to a class that were correctly identified by the model. F1-Score represents the harmonic mean of Precision and Recall and provides a balanced measure of classification performance.

Macro F1-score was particularly important because it assigns equal importance to each of the three outcome classes. This is relevant because the dataset is not perfectly balanced, with Graduate representing the largest class and Enrolled the smallest.

Confusion Matrices were also used to examine the specific classification errors made by each model. The use of multiple evaluation metrics was necessary because overall accuracy alone does not fully describe the performance of a multiclass classifier. In particular, the results showed that prediction of the Enrolled class was substantially more difficult than prediction of the Graduate and Dropout classes.

4.3 Model Performance Comparison

The results obtained from the testing dataset are summarized below.

Model	Accuracy	Dropout F1	Enrolled F1	Graduate F1	Macro F1
Logistic Regression	0.77	0.78	0.44	0.86	0.69
Decision Tree	0.66	0.68	0.33	0.78	0.60
Random Forest	0.77	0.78	0.47	0.85	0.70
XGBoost	0.76	0.77	0.45	0.85	0.69

The results indicate that Random Forest achieved the strongest overall performance among the four models. Although its accuracy of approximately 0.77 was tied with Logistic Regression, Random Forest achieved the highest macro F1-score at approximately 0.70. This indicates that it provided the most balanced performance across the three target categories.

Logistic Regression also performed strongly, achieving approximately 0.77 accuracy and a 0.69 macro F1-score. XGBoost achieved approximately 0.76 accuracy and a 0.69 macro F1-score, while Decision Tree produced the lowest performance with approximately 0.66 accuracy and a 0.60 macro F1-score.

The results demonstrate that model selection should not be based solely on accuracy. The slightly higher macro F1-score of Random Forest indicates better overall balance across the three classes.

4.4 Class-Specific Model Performance

An important finding from the model evaluation is the difference in predictive performance among the three target categories.

The Graduate class was consistently the easiest class for the models to identify. Random Forest achieved an F1-score of approximately 0.85 for Graduate, while Logistic Regression and XGBoost achieved approximately 0.86 and 0.85, respectively.

The Dropout class was also predicted relatively well. Random Forest achieved an F1-score of approximately 0.78, indicating that the model was reasonably effective in identifying students associated with the dropout outcome.

In contrast, the Enrolled class presented the greatest predictive challenge. Random Forest achieved an F1-score of only approximately 0.47, compared with 0.44 for Logistic Regression, 0.33 for Decision Tree, and 0.45 for XGBoost.

This result is important from an institutional perspective. The Enrolled category represents students who have not yet reached either the Graduate or Dropout outcome. Their characteristics may therefore overlap with both groups, making them more difficult to distinguish using the available variables.

The relatively low performance for Enrolled demonstrates why macro F1-score and class-specific evaluation are important in this project. A model that achieves high accuracy primarily through strong Graduate classification may not necessarily provide equally useful predictions across all student outcomes.

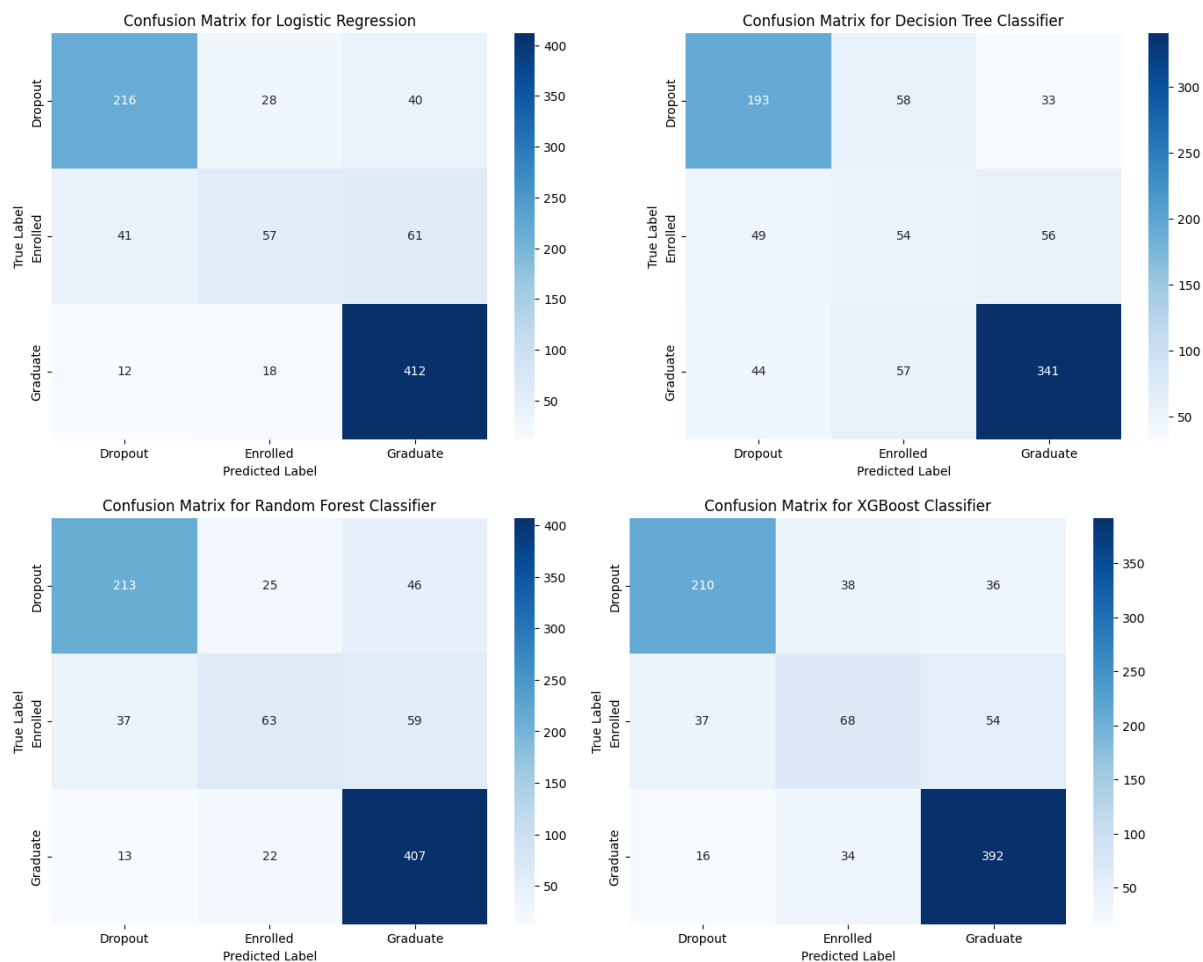
4.5 Confusion Matrix Analysis

Confusion matrices were generated to examine the classification errors produced by the models.

The confusion matrices provide information regarding whether students were correctly classified as Dropout, Enrolled, or Graduate and identify instances in which the models confused one outcome category with another.

The results indicate that the models were substantially more successful in identifying Graduate students than Enrolled students. This pattern is consistent with the class-specific F1-scores and suggests that the characteristics of students who remain enrolled may overlap with those of students who eventually graduate or drop out.

The Random Forest confusion matrix should receive particular attention because Random Forest was selected as the final model. The matrix provides a more detailed understanding of the model's strengths and limitations than accuracy alone.



4.6 Model Selection

Based on the comparative evaluation, Random Forest was selected as the preferred model for the project.

The selection was based primarily on its highest macro F1-score of approximately 0.70, combined with an overall accuracy of approximately 0.77. The model also achieved the highest F1-score among the evaluated models for the Enrolled category, although performance for this class remained relatively low.

Random Forest also provides an important interpretability advantage because its feature-importance scores can be used to identify the variables contributing most strongly to the classification process. This characteristic was particularly valuable for the subsequent Explainable AI analysis.

Therefore, Random Forest provided the strongest combination of predictive performance and interpretability for the objectives of this capstone.

4.7 Feature Importance

Feature-importance analysis identified several variables that contributed substantially to the Random Forest classification model.

The most influential predictors were primarily related to students' academic performance during the first and second semesters. Important predictors included Curricular Units – 2nd Semester Grade, Curricular Units – 2nd Semester Approved, Curricular Units – 1st Semester Grade, Curricular Units – 1st Semester Approved, Tuition Fees Up to Date, Scholarship Holder, Admission Grade, and Age at Enrollment.

The dominance of first- and second-semester academic variables indicates that early academic performance contains substantial predictive information regarding students' eventual academic outcomes.

Financial variables also contributed to prediction. Tuition payment status and scholarship status were identified as important predictors, suggesting that financial circumstances may be associated with students' academic trajectories.

4.8 Chapter Summary

This chapter presented the implementation and evaluation of four supervised machine learning algorithms for predicting student academic outcomes: Logistic Regression, Decision Tree, Random Forest, and XGBoost.

The models were evaluated using the independent testing dataset and compared using Accuracy, Precision, Recall, F1-Score, macro F1-score, and Confusion Matrix analysis. The results showed that Random Forest achieved the strongest overall performance, with approximately 0.77 accuracy and 0.70 macro F1-score.

The analysis also revealed an important limitation: while the models performed relatively well in identifying Graduate and Dropout students, prediction of the Enrolled class remained considerably more difficult. This finding demonstrates the importance of using class-specific evaluation metrics rather than relying solely on overall accuracy.

Random Forest was therefore selected as the preferred model because it provided the strongest balance of predictive performance across the three outcome categories and offered useful feature-importance information for interpretation.

Chapter 5

Critical Thinking – Ethical AI and Bias Auditing

5.1 Introduction

The final stage of the project focused on interpreting the machine learning results and assessing the developed predictive models from the perspectives of explainability, limitations, fairness, and ethical use. While predictive accuracy is an important consideration in machine learning, a model intended to support educational decision-making must also provide understandable and responsible predictions. In the context of student academic outcomes, an incorrect prediction may influence how educators identify students for academic advising, mentoring, financial assistance, or other forms of intervention. Therefore, the model should not be treated as an automated decision-maker but as a decision-support tool that complements professional judgment.

The modeling stage compared four supervised machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Random Forest provided the strongest balance of predictive performance and interpretability and was therefore used extensively for feature-importance and explainability analysis.

5.2 Model Explainability

Explainable Artificial Intelligence (XAI) techniques were incorporated to improve the transparency and interpretability of the machine learning models. The primary purpose of explainability was to determine which student characteristics contributed to predictions of Dropout, Enrolled, or Graduate and to provide both global and individual-level interpretations of model behavior.

The analysis used several complementary explainability approaches, including feature importance, SHAP (SHapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations), Partial Dependence Plots (PDP), and Individual Conditional Expectation (ICE) plots. These techniques provide different perspectives

on model behavior. Global methods help identify variables that are generally influential across the dataset, while local methods explain the prediction of an individual student.

5.2.1 Feature Importance

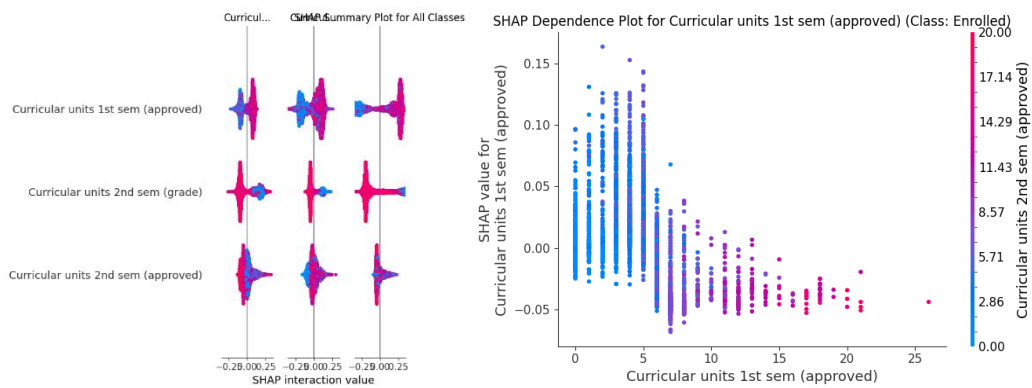
Feature-importance analysis identified academic performance variables as the strongest predictors of student outcomes. The most influential variables included Curricular Units Approved and Curricular Units Grade for both the first and second semesters. Other important variables included Tuition Fees Up to Date, Scholarship Holder, Admission Grade, and Age at Enrollment.

The dominance of first- and second-semester academic indicators suggests that academic progression is strongly associated with a student's eventual educational outcome. In practical terms, students who demonstrate stronger academic progress are more likely to be classified as Graduate, whereas students with weaker academic performance are more likely to be classified as Dropout.

This finding is particularly relevant to the business problem because academic performance indicators can potentially serve as early warning signals. Institutions may therefore use these indicators to prioritize students for academic support before difficulties become severe.

5.2.2 SHAP Analysis

SHAP was used to provide a more detailed interpretation of how individual features contribute to model predictions. SHAP values provide information about the direction and magnitude of a feature's contribution to a prediction. This allows the analysis to move beyond simply identifying important variables and examine how specific feature values influence the predicted outcome.



The SHAP analysis reinforced the importance of academic performance variables. Curricular Units Approved, semester grades, Tuition Fees Up to Date, and Scholarship Holder were among the variables identified as important to student-outcome prediction. Higher academic achievement was generally associated with stronger predictions toward graduation, while weaker academic performance was associated with greater likelihood of dropout.

These findings provide an important connection between model performance and educational interpretation. The model is not simply identifying statistical patterns without context; its most influential variables correspond to measurable indicators of academic progress that educational institutions can monitor.

5.2.3 LIME Analysis

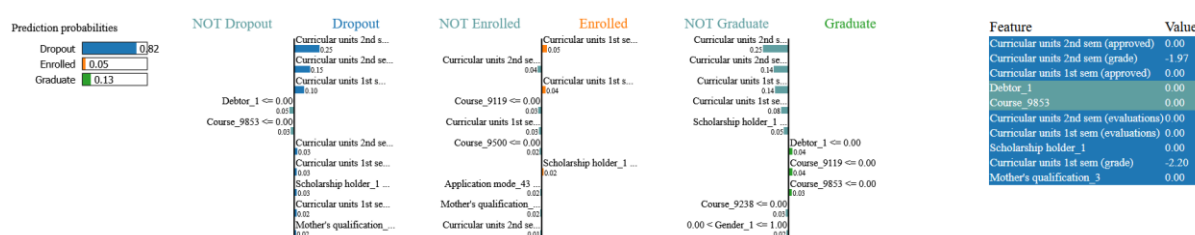
Local Interpretable Model-Agnostic Explanations (LIME) was implemented to examine individual student predictions generated by the Random Forest model. Unlike global feature-importance and SHAP analyses, LIME focuses on a specific observation and identifies the features that most strongly contributed to its classification.

Three representative cases were examined: one Dropout case, one Enrolled case, and one Graduate case. The LIME outputs generated probability distributions for all three target classes and identified the ten most influential local features for each prediction.

5.2.3.1 Dropout Prediction

For the selected Dropout case, the Random Forest model assigned an 82% probability of Dropout, compared with 5% for Enrolled and 13% for Graduate. The strongest factors supporting the Dropout prediction were primarily related to academic performance. The largest positive contribution came from having two or fewer approved curricular units in the second semester, followed by a very low second-semester grade and having three or fewer approved curricular units during the first semester. Other contributing variables included the number of evaluations and first-semester academic performance.

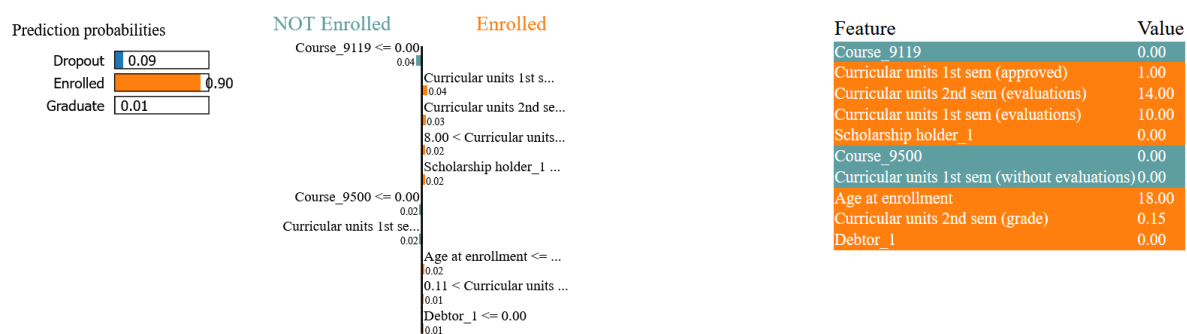
These results indicate that poor academic progression was the dominant local pattern supporting the Dropout prediction. The LIME explanation therefore provides a student-level interpretation of the broader feature-importance results: students demonstrating weak academic progress are more likely to be classified as Dropout.



5.2.3.2 Enrolled Prediction

For the selected Enrolled case, the model assigned a 90% probability of Enrolled, with 9% probability of Dropout and 1% probability of Graduate. The LIME explanation showed that the prediction was influenced by a combination of academic-performance indicators, curricular-unit evaluations, scholarship status, age at enrollment, and course-related variables. Unlike the Dropout case, the Enrolled prediction was not dominated by extremely weak academic indicators. Instead, several characteristics collectively contributed to the classification.

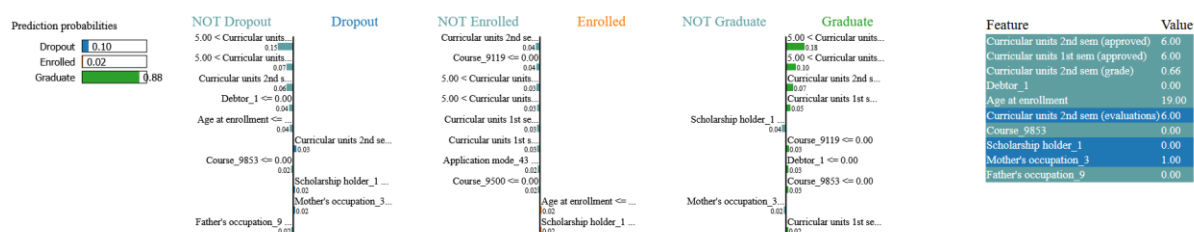
This observation is important because the Enrolled category represents students whose eventual academic outcome has not yet been resolved in the original dataset. Students in this category may exhibit characteristics that overlap with both successful and unsuccessful students. Consequently, distinguishing Enrolled students from the other two outcomes may be inherently more challenging.



5.2.3.3 Graduate Prediction

For the selected Graduate case, the model assigned an 88% probability of Graduate, compared with 10% for Dropout and 2% for Enrolled. The Graduate-specific LIME explanation showed that stronger academic progression was the most important contributor to the prediction. The strongest positive contribution came from having between five and six approved curricular units during the second semester. This was followed by the corresponding first-semester approved-unit range, a second-semester grade above 0.60, and a first-semester grade above 0.57.

The result provides a clear contrast with the Dropout case. While low numbers of approved curricular units and low grades pushed the model toward Dropout, stronger academic progression and higher grades pushed the model toward Graduate.



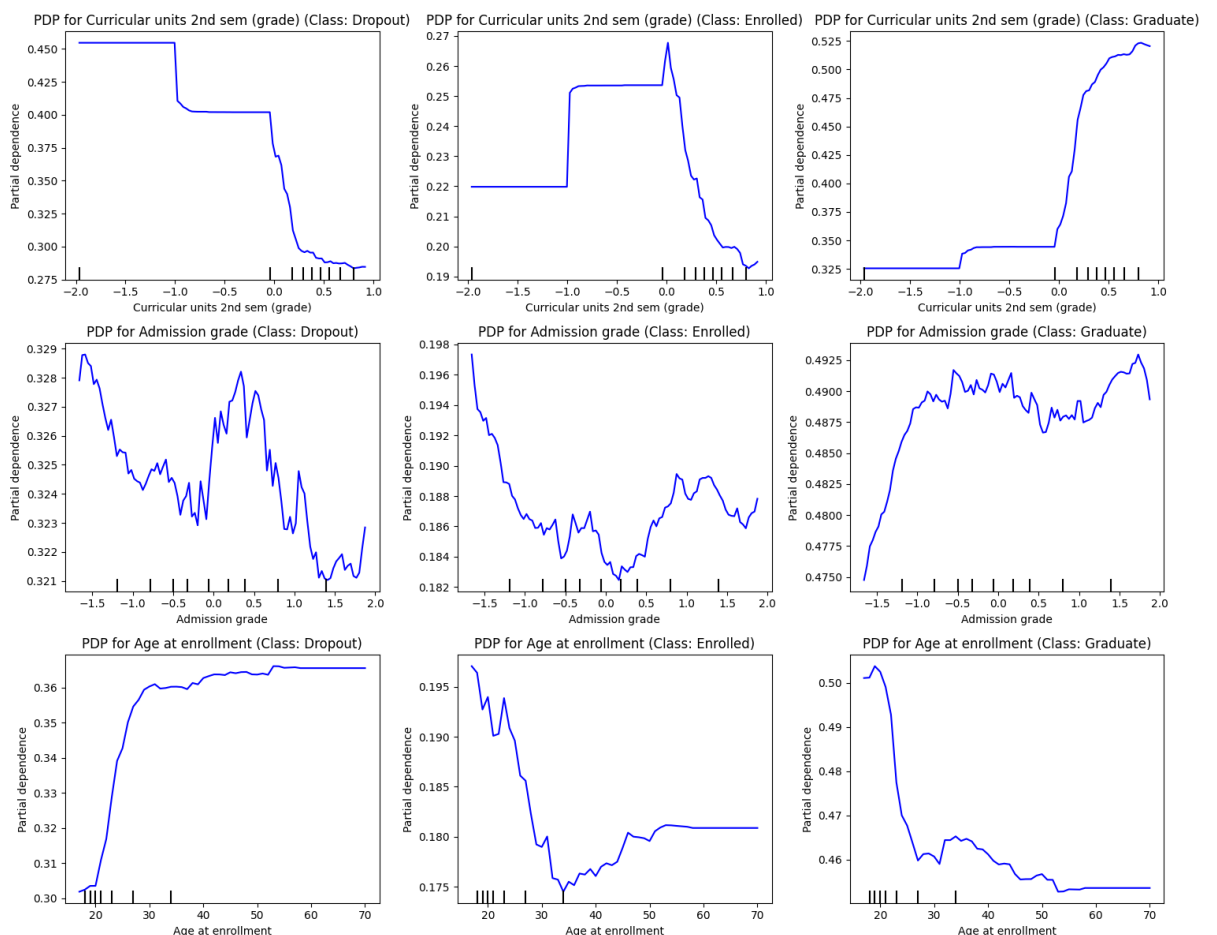
LIME Analysis Summary

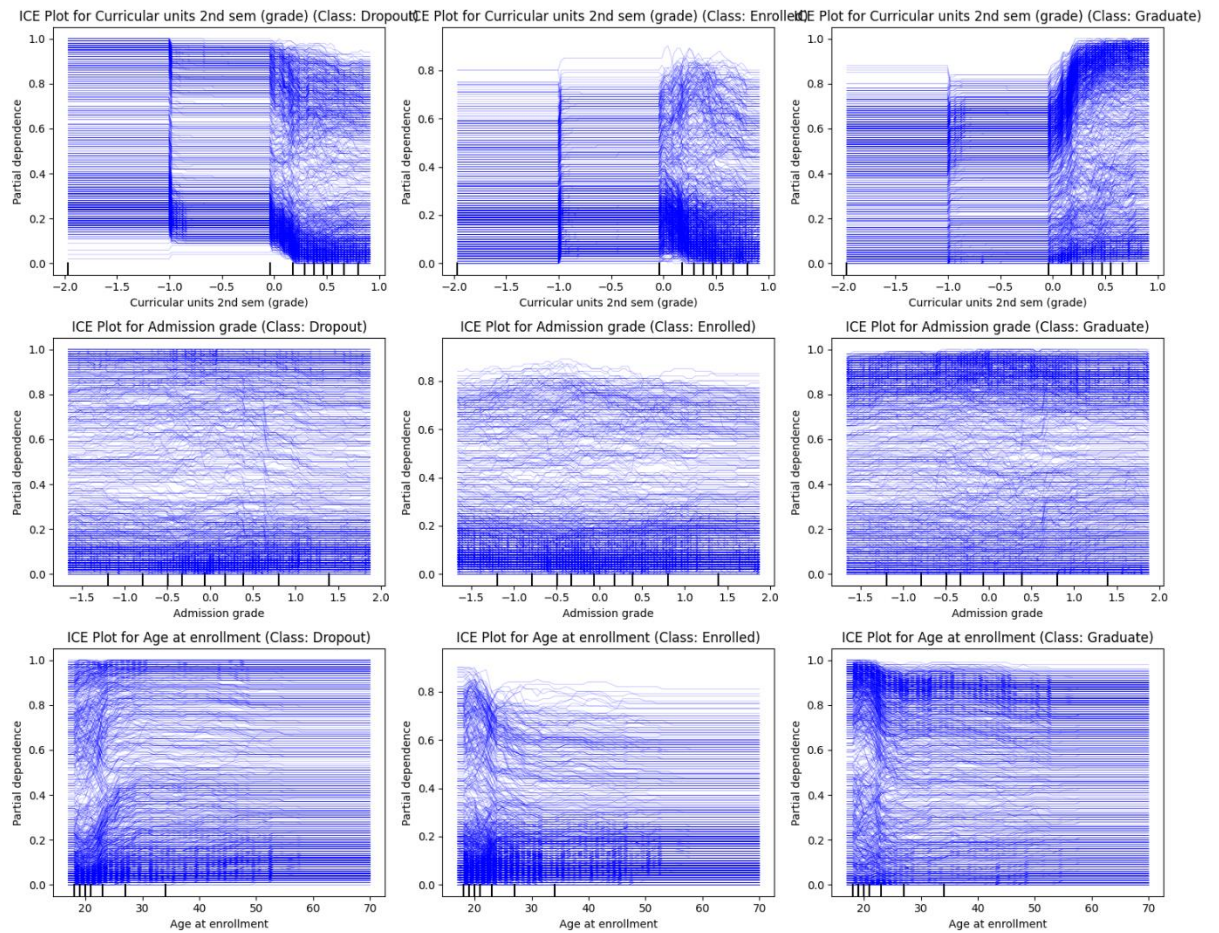
Overall, the LIME analysis provides local evidence supporting the global feature-importance findings. The Dropout case was primarily characterized by weak academic progression, while the Graduate case was associated with stronger academic achievement and curricular-unit completion. The Enrolled case demonstrated a more complex combination of academic and student characteristics.

An important methodological consideration is that each LIME HTML output contains explanations for all three target classes. Therefore, the explanation corresponding to the predicted class must be used when interpreting each case. For the three-class model, Dropout corresponds to class 0, Enrolled corresponds to class 1, and Graduate corresponds to class 2. The Graduate interpretation should use the class-2 explanation rather than simply the first explanation displayed in the HTML file.

5.3 Partial Dependence and Individual Conditional Expectation Analysis

Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots were used to further investigate the relationship between important predictor variables and model predictions. PDP provides an estimate of the average relationship between a feature and the model's prediction, while ICE examines how that relationship varies for individual observations.





The analysis indicated that the number of approved curricular units had a strong relationship with predicted student outcomes. Increasing academic progression was generally associated with an increased probability of Graduate and a reduced probability of Dropout.

The ICE analysis further demonstrated that individual students do not necessarily respond to changes in a predictor in exactly the same way. Differences in students' overall profiles can result in different prediction patterns. This finding reinforces the importance of avoiding overly simplistic interpretations of individual predictors.

Together, PDP and ICE provide a more nuanced interpretation of the model. They demonstrate that while academic progression is an important general predictor, student outcomes are influenced by combinations of characteristics rather than by a single variable in isolation.

5.4 Critical Assessment of Model Limitations

Despite the predictive performance achieved by the developed models, several limitations should be considered when interpreting the results.

First, the target variable contains three outcome categories with unequal representation. Class imbalance can cause a classifier to favor the more frequently represented classes and may reduce its ability to identify less frequent outcomes accurately. This limitation is particularly important when the practical objective is to identify students at risk of dropping out.

Second, the model's performance depends on the variables available in the dataset. Some of the strongest predictors are first- and second-semester academic indicators. While these variables are highly informative, their availability may limit how early the model can be used for intervention. A model intended for early-warning purposes should ideally rely on information available sufficiently early in the student's academic journey.

Third, ensemble models such as Random Forest and XGBoost are more complex than simple statistical models. Although they can capture nonlinear relationships and interactions between variables, their complexity makes direct interpretation more difficult. XAI techniques therefore become important for understanding their predictions.

Finally, the dataset represents a particular educational context. Differences in admission policies, academic programs, student demographics, grading practices, and institutional support systems may affect model performance when the model is applied to another institution. External validation using data from other educational institutions would therefore be necessary before operational deployment.

5.5 Bias Detection and Fairness Auditing

Because the model may be used to identify students who require additional support, fairness is an important consideration. A predictive system should not

systematically disadvantage students because of demographic or socioeconomic characteristics.

The fairness assessment considered variables available in the dataset that may represent sensitive or potentially sensitive characteristics. These included Gender, Age at Enrollment, Scholarship Holder, Debtor Status, and International Student Status. The purpose of examining these attributes was to determine whether predictions could systematically differ across student groups.

Three fairness concepts were considered: Demographic Parity, Equalized Odds, and Disparate Impact. Demographic Parity examines differences in prediction rates between groups. Equalized Odds considers whether true-positive and false-positive rates are comparable across groups. Disparate Impact evaluates whether the rate at which a particular outcome is assigned differs substantially between groups.

The analysis suggests that academic-performance variables were more prominent in the model's predictions than demographic characteristics. However, this does not eliminate the possibility of indirect socioeconomic bias. Variables such as tuition-payment status and scholarship status may reflect underlying financial circumstances that can themselves be associated with educational achievement.

Therefore, a model may appear to rely primarily on academic variables while still indirectly reflecting socioeconomic differences. This is an important ethical consideration when using predictive analytics in education.

5.6 Bias Mitigation Strategies

Several strategies can be considered to reduce potential bias and improve the robustness of the predictive system.

One approach is class reweighting, in which greater importance is assigned to underrepresented target classes during model training. This can reduce the tendency of a classifier to favor majority outcomes. Another potential strategy is threshold

optimization, which can be used to adjust classification decisions when disparities in prediction errors are identified.

Data-level approaches such as oversampling can also be considered when class imbalance affects predictive performance. Techniques such as SMOTE may increase the representation of underrepresented classes; however, such techniques should be applied carefully to avoid creating unrealistic synthetic observations.

Fairness should also be monitored continuously rather than assessed only once during model development. As new student cohorts are introduced, changes in demographic composition, academic policies, economic conditions, and institutional practices may affect model behavior. Periodic retraining and fairness auditing can therefore help maintain model reliability over time.

5.7 Ethical Considerations

The ethical use of the predictive model is particularly important because its predictions concern students' educational outcomes. The model should therefore function as a decision-support system rather than a decision-making authority.

A prediction of Dropout should not be interpreted as a deterministic statement that a student will leave the institution. Instead, it should be interpreted as an indication that the student may benefit from additional assessment or support. Educators should consider the student's circumstances and use professional judgment before taking action.

The model should be used primarily to facilitate supportive interventions such as academic advising, tutoring, mentoring, counseling, and financial assistance. Students should not be penalized, excluded, or subjected to adverse decisions solely because they receive a high predicted probability of Dropout.

Student privacy is another major consideration. Student records may contain personal, financial, demographic, and academic information. Consequently, access to

the data and model predictions should be appropriately controlled, and student information should be protected throughout the data lifecycle.

Transparency is also essential. Through SHAP, LIME, PDP, and ICE, educators can obtain explanations of model behavior rather than relying entirely on unexplained predictions. This improves accountability and allows questionable predictions to be reviewed by human decision-makers.

5.8 Critical Thinking and Practical Implications

The findings of this project demonstrate that high predictive performance alone is insufficient for determining whether a machine learning model is appropriate for educational use. A model may achieve high accuracy while still producing undesirable outcomes for particular groups or failing to detect an important minority class.

The explainability analysis provides an important additional perspective. Across the global and local analyses, academic progression emerged as a dominant factor associated with student outcomes. In the LIME analysis, the selected Dropout student had very low numbers of approved curricular units and low grades, whereas the selected Graduate student demonstrated stronger curricular-unit completion and semester grades. These results provide a logical connection between the model's statistical behavior and recognizable academic indicators.

However, the results should not be interpreted as demonstrating that academic performance alone causes dropout or graduation. Machine learning identifies predictive relationships rather than establishing causal relationships. Academic performance may itself be influenced by other factors, including financial circumstances, personal circumstances, institutional support, and other variables that may not be fully captured in the dataset.

This distinction is particularly important when translating the model into institutional practice. The appropriate response to a high-risk prediction is not to label a student as likely to fail, but to investigate the underlying circumstances and determine whether appropriate support can be provided.

5.9 Overall Chapter Summary

This chapter evaluated the developed machine learning framework from the perspectives of explainability, limitations, fairness, and ethical use. The analysis demonstrated that Explainable AI techniques can provide valuable insight into the behavior of complex machine learning models.

Feature importance and SHAP analysis identified academic-performance variables, particularly approved curricular units and semester grades, as major predictors of student outcomes. LIME provided individual-level explanations and demonstrated that weak academic progression strongly contributed to the selected Dropout prediction, while stronger academic progression contributed to the selected Graduate prediction. The selected Enrolled case demonstrated a more complex combination of academic and student characteristics.

PDP and ICE analyses further demonstrated the relationship between academic progression and model predictions while highlighting differences among individual student profiles. These explainability methods collectively provide a stronger basis for interpreting the model than predictive performance metrics alone.

The chapter also identified important limitations, including class imbalance, model complexity, the timing and availability of academic predictors, and limited generalizability beyond the original educational context. Fairness considerations highlighted the possibility of indirect socioeconomic bias even when demographic variables are not the dominant predictors.

The developed model should be viewed as an early-warning and decision-support mechanism rather than an automated student classification system. Its greatest practical value lies in helping educators identify students who may benefit from timely support while preserving human judgment, student privacy, fairness, and accountability. This responsible approach ensures that predictive analytics contributes to improved student retention and academic success without turning predictions into predetermined educational outcomes.

Chapter 6

Overall Conclusion and Recommendations

This study successfully developed and evaluated a machine learning framework for predicting student academic outcomes using the Students Dropout and Academic Success dataset. The primary objective was to determine whether students could be classified as Dropout, Enrolled, or Graduate based on demographic, socioeconomic, financial, and academic characteristics. Through systematic data preparation, exploratory data analysis, feature selection, machine learning model development, model evaluation, and Explainable Artificial Intelligence (XAI) analysis, the study demonstrated the potential of machine learning as a decision-support approach for improving student retention and academic success.

The data understanding and preprocessing stages established that the dataset was suitable for machine learning analysis. The dataset contained no missing values or duplicate records, reducing the need for extensive data-cleaning procedures. Exploratory analysis showed meaningful differences in academic characteristics across the three student outcome categories. In particular, variables associated with curricular-unit completion and semester academic performance demonstrated strong relationships with students' eventual outcomes.

Feature-selection analysis further identified academic performance as a major source of predictive information. Variables related to the number of curricular units approved, semester grades, tuition payment status, scholarship status, admission grade, and age at enrollment emerged as important predictors. The prominence of first- and second-semester academic indicators suggests that academic progression provides valuable information for distinguishing students who eventually graduate from those who remain enrolled or drop out.

Four supervised machine learning algorithms were implemented and evaluated: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The models were compared using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis. The results showed that ensemble-based approaches provided competitive predictive performance, while the Decision Tree produced comparatively

weaker results. Among the evaluated models, Random Forest achieved the strongest overall balance of performance, with approximately 0.77 test accuracy and a 0.70 macro F1-score. Logistic Regression and XGBoost also demonstrated competitive performance, while Decision Tree achieved the lowest overall results.

An important finding from the evaluation was the difference in predictive performance across the three outcome categories. The models performed relatively well in identifying Graduate and Dropout students but experienced greater difficulty in predicting the Enrolled class. This finding suggests that students who remain enrolled may possess characteristics that overlap with both eventual graduates and dropouts. Consequently, overall accuracy alone does not fully represent the usefulness of the model, and class-specific measures such as macro F1-score and confusion matrices are important when assessing its effectiveness.

The study also incorporated Explainable Artificial Intelligence techniques to improve the transparency of the selected Random Forest model. Feature importance, SHAP, LIME, Partial Dependence Plots, and Individual Conditional Expectation analysis were used to examine the factors influencing model predictions. The results consistently emphasized the importance of academic progression. The LIME analysis provided additional individual-level evidence: the selected Dropout student received an 82% predicted probability of Dropout, with low approved curricular units and weak semester grades contributing strongly to the prediction. In contrast, the selected Graduate student received an 88% probability of Graduate, with stronger curricular-unit completion and semester grades contributing positively to the prediction. The selected Enrolled student received a 90% probability of Enrolled, with several academic and student characteristics contributing collectively to the prediction.

These findings demonstrate the value of combining predictive modeling with explainability. Rather than simply producing a classification, the model can provide information about the factors associated with an individual prediction. This is particularly important in an educational setting where predictions should be understandable to educators and used to support—not replace—professional judgment.

The study also considered fairness and ethical issues associated with the use of predictive analytics in education. Although academic-performance variables were more prominent in the model's explanations, the presence of financial and socioeconomic indicators such as tuition status and scholarship status indicates the possibility of indirect socioeconomic influence. Consequently, predictive models used for student intervention should be subjected to continuing fairness assessment, privacy protection, and human oversight.

Overall, the findings demonstrate that machine learning can provide useful predictive information regarding student academic outcomes. However, the model should not be interpreted as determining a student's future with certainty. Instead, it should function as an early-warning and decision-support mechanism that helps educators identify students who may require additional academic, financial, or personal support.

Recommendations

Based on the findings of the study, the following recommendations are proposed:

1. **Develop an Early-Warning System.**
Integrate the predictive model into student-support processes to identify students who may require timely academic, financial, or personal assistance.
2. **Monitor Academic Progression.**
Give particular attention to curricular units approved and semester grades, which were identified as among the strongest predictors of student outcomes.
3. **Improve Prediction of Enrolled Students.**
Future studies should focus on improving prediction of the **Enrolled** class, which demonstrated the weakest classification performance. Additional indicators such as attendance, engagement, and learning activity may be considered.
4. **Continuously Validate the Model.**
Retrain and evaluate the model using updated student data to maintain its accuracy and relevance as student populations and educational conditions change.
5. **Maintain Explainability and Fairness.**
Continue using XAI techniques such as **SHAP and LIME** and conduct periodic fairness assessments to ensure that predictions are transparent and equitable.
6. **Ensure Human Oversight and Ethical Use.**
The model should serve as a **decision-support tool rather than a replacement for professional judgment**. Predictions should be used to initiate appropriate student support rather than to make punitive or automatic decisions.

Final Remarks

This study demonstrates the potential of machine learning to support data-driven decision-making in education. By integrating data preprocessing, exploratory analysis, feature selection, supervised machine learning, model evaluation, and Explainable AI, the project established a framework for predicting student outcomes and examining the factors underlying those predictions.

The findings indicate that academic progression—particularly curricular-unit completion and semester performance—is strongly associated with predicted student outcomes. The Random Forest model provided the strongest overall balance of predictive performance among the evaluated models, while LIME and other XAI techniques demonstrated how the model's predictions could be interpreted at both global and individual levels.

At the same time, the study recognizes that predictive accuracy does not establish causality and that machine learning predictions contain uncertainty. The relatively weaker prediction of the Enrolled category also demonstrates that student outcomes are complex and cannot always be distinguished through available historical variables alone.

The practical contribution of this study therefore lies not in replacing human judgment but in augmenting it with evidence derived from student data. When implemented responsibly, the predictive framework can help educators identify students who may benefit from timely intervention, allowing institutions to move from primarily reactive approaches toward more proactive and evidence-based student support.

The responsible application of machine learning in education should focus not on predicting failure, but on using prediction to enable timely support, improve student retention, promote academic success, and create a more responsive and supportive learning environment.

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